

STAT 5680

Statistical Collaboration

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	AI Engineer (92.2 / 100)	80.3 / 100

80

CURRENT AVG SCORE
across 6 job roles

100

POTENTIAL SCORE
with recommended additions

+35

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Educates and trains students to become effective interdisciplinary collaborators by developing the communication and collaboration skills necessary to apply technical statistics and data science skills to help domain experts answer research questions. Topics include structuring effective meetings and projects; communicating statistics to non-statisticians; using peer feedback, self-reflection and video analysis to improve collaboration skills; creating reproducible statistical workflows; working ethically.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	92.2	100.0	+7.8	LOW
Applied Scientist	89.4	100.0	+10.6	LOW
ML Engineer	82.2	100.0	+17.8	MEDIUM
Machine Learning Engineer	79.1	100.0	+20.9	MEDIUM
Data Scientist	73.8	97.1	+23.3	MEDIUM
MLOps Engineer	65.3	100.0	+34.7	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1 LOW Optional: Business Communication — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

AI Engineer 92.2 → 100.0 (+7.8 pts)

models learning software model teams problems machine design

Applied Scientist 89.4 → 100.0 (+10.6 pts)

computational reasoning problems models department learning industry design

ML Engineer 82.2 → 100.0 (+17.8 pts)

learning machine models model world problems new building

Machine Learning Engineer 79.1 → 100.0 (+20.9 pts)

learning tools business design development models model applications

Data Scientist 73.8 → 97.1 (+23.3 pts)

toolkit presentation problems

MLOps Engineer 65.3 → 100.0 (+34.7 pts)

model scientists learning machine tools models development

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.